



HEALTHCARE EQUIPMENT SYSTEM

A Smart Backend Solution for Managing Medical Equipment Efficiently

Spring Boot

MySQL

JPA / Hibernate

REST API

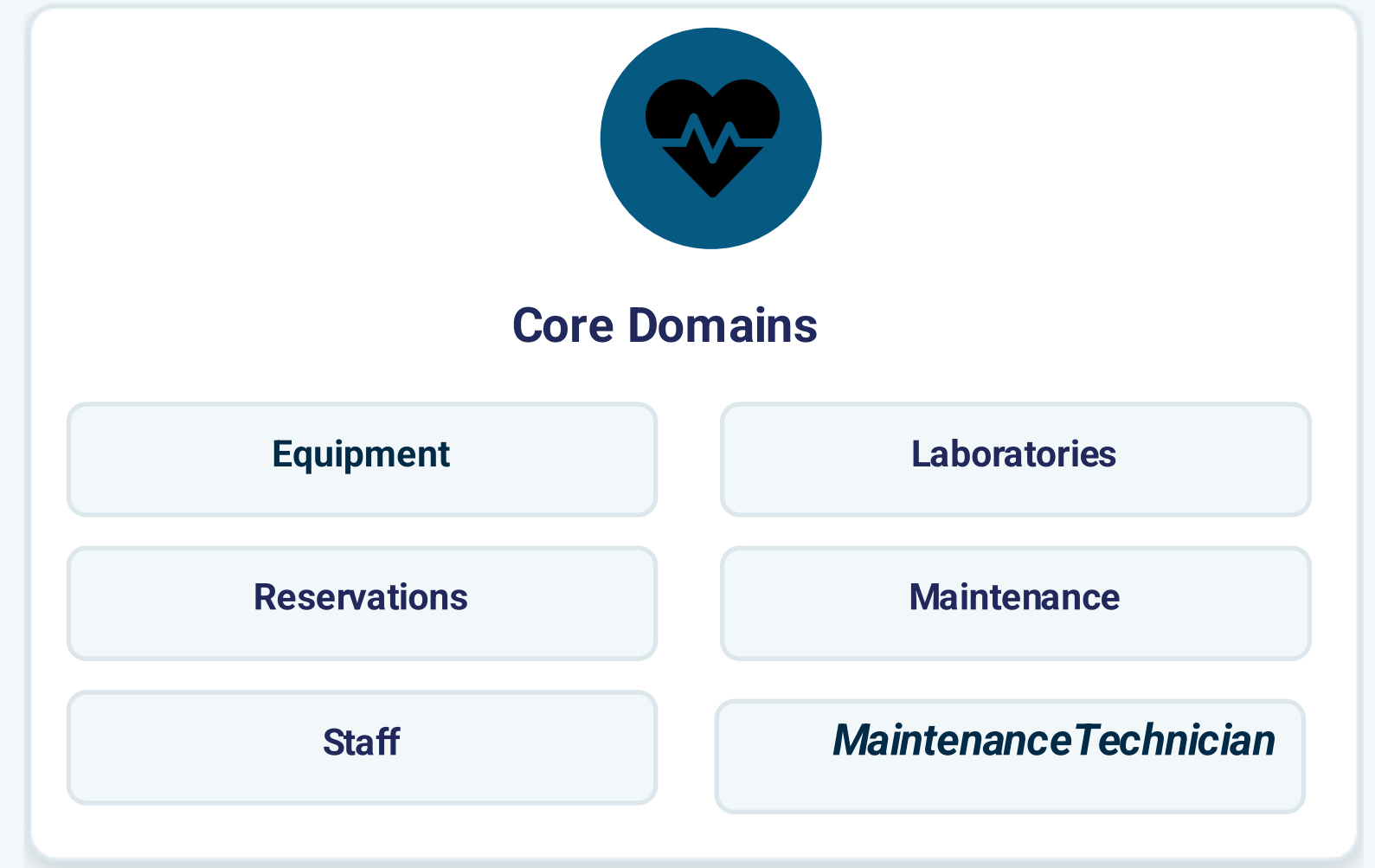
Docker

Backend Application · Equipment · Laboratory · Reservation · Maintenance Management

Overview

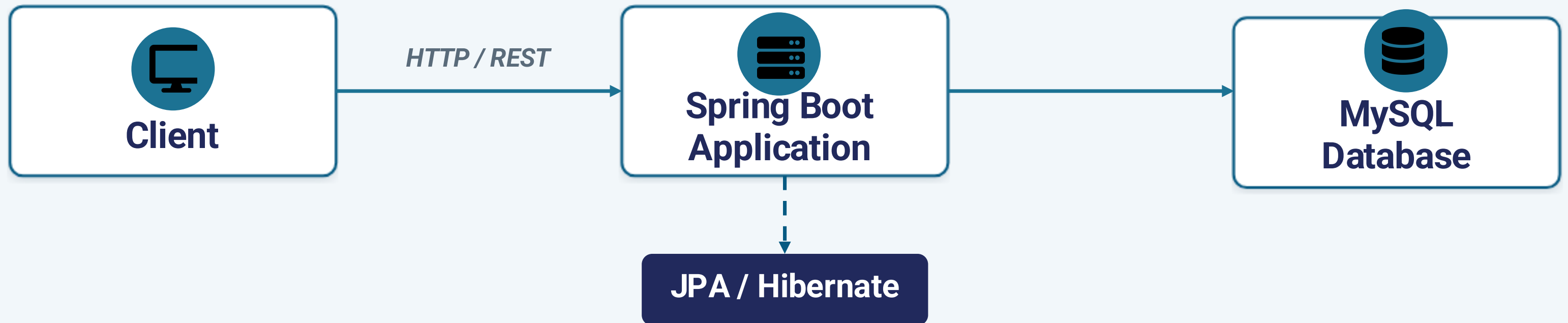
The Healthcare Equipment System is a backend application built with Spring Boot to manage medical equipment, laboratories, staff, reservations, and maintenance.

- ✓ Manage equipment and laboratories
- ✓ Handle reservations and availability
- ✓ Track maintenance history
- ✓ Role-based access for staff
- ✓ RESTful APIs for integration



System Architecture

Three-tier request flow



Client sends HTTP requests to the Spring Boot application, which uses JPA/Hibernate to persist and retrieve data from a MySQL database.

Key Features

What the system delivers



Equipment Management

Add, update, view and deactivate equipment



Laboratory Management

Manage laboratories and their equipment



Reservation Management

Reserve equipment and check availability



Maintenance Tracking

Record maintenance and track history



Soft Delete (isActive)

Deactivate records instead of deleting



RESTful API

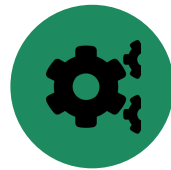
Well-structured APIs for front-end & systems

Technology Stack

Tools and technologies used



Java 17



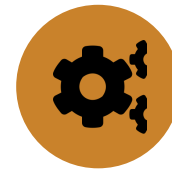
Spring Boot 3.x



Spring Data JPA



MySQL 8.0



Maven

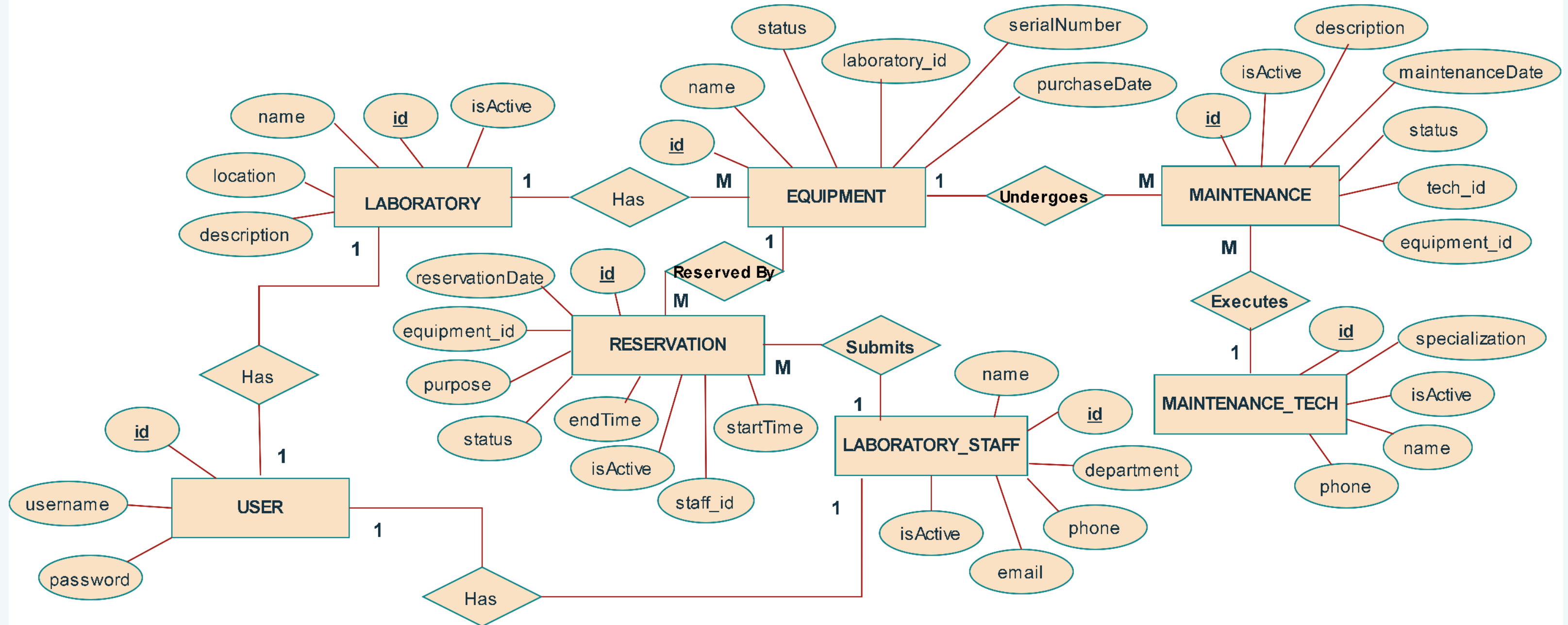


Docker

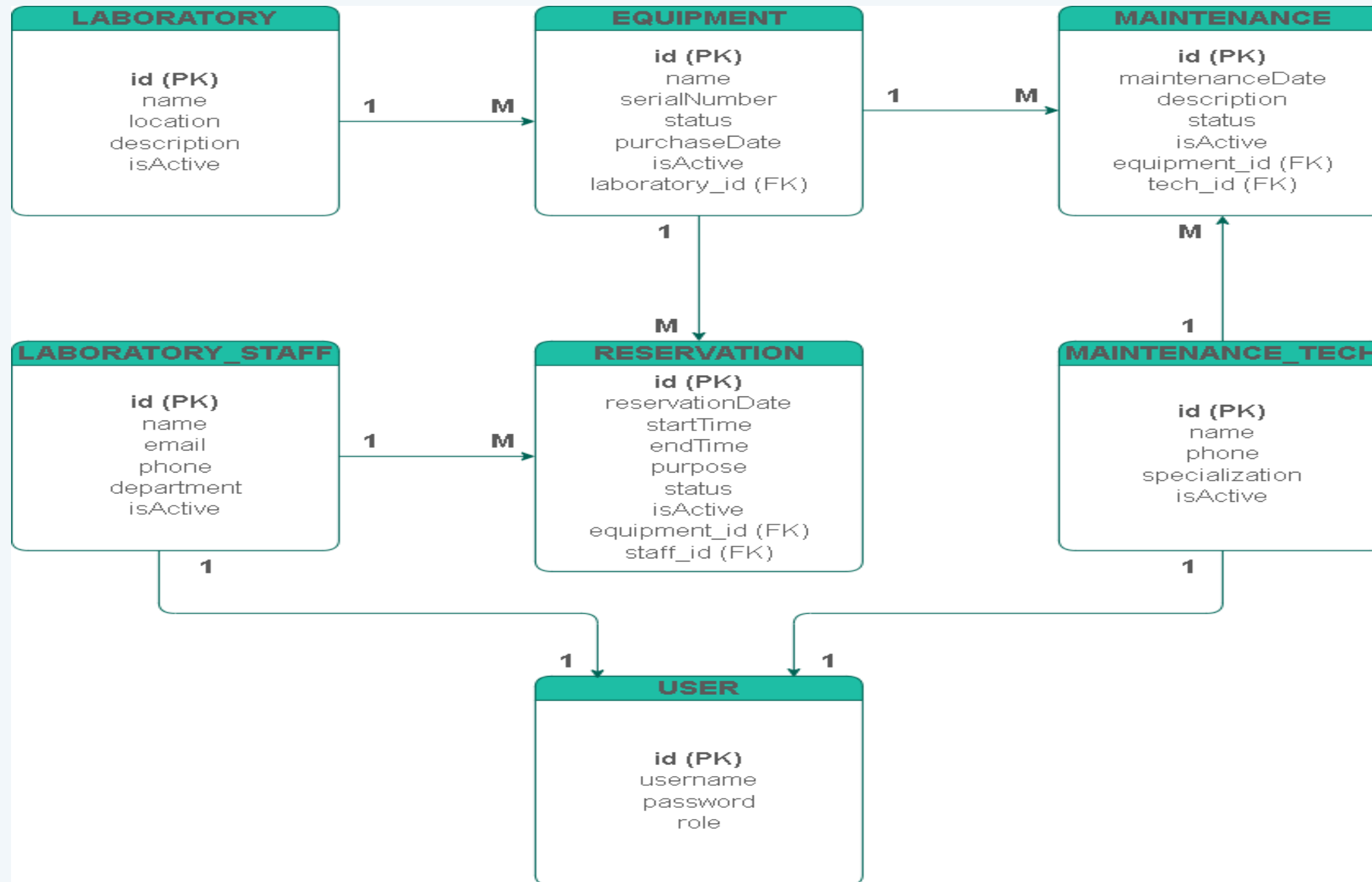
Supporting tools

- IntelliJ IDEA
- Postman
- MySQL Workbench
- Git & GitHub

ERD – Entity Relationship Diagram



Relational Schema for Database



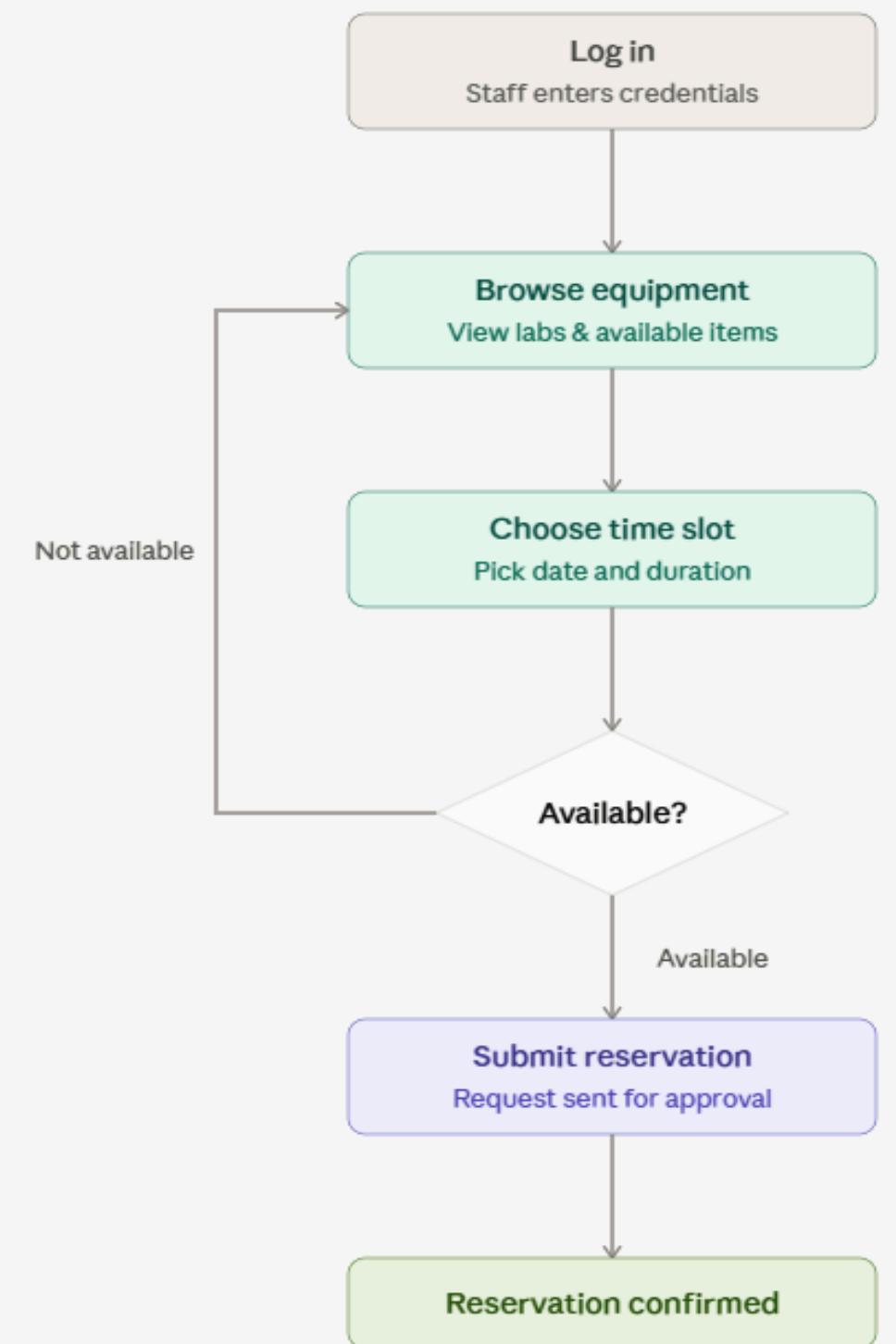
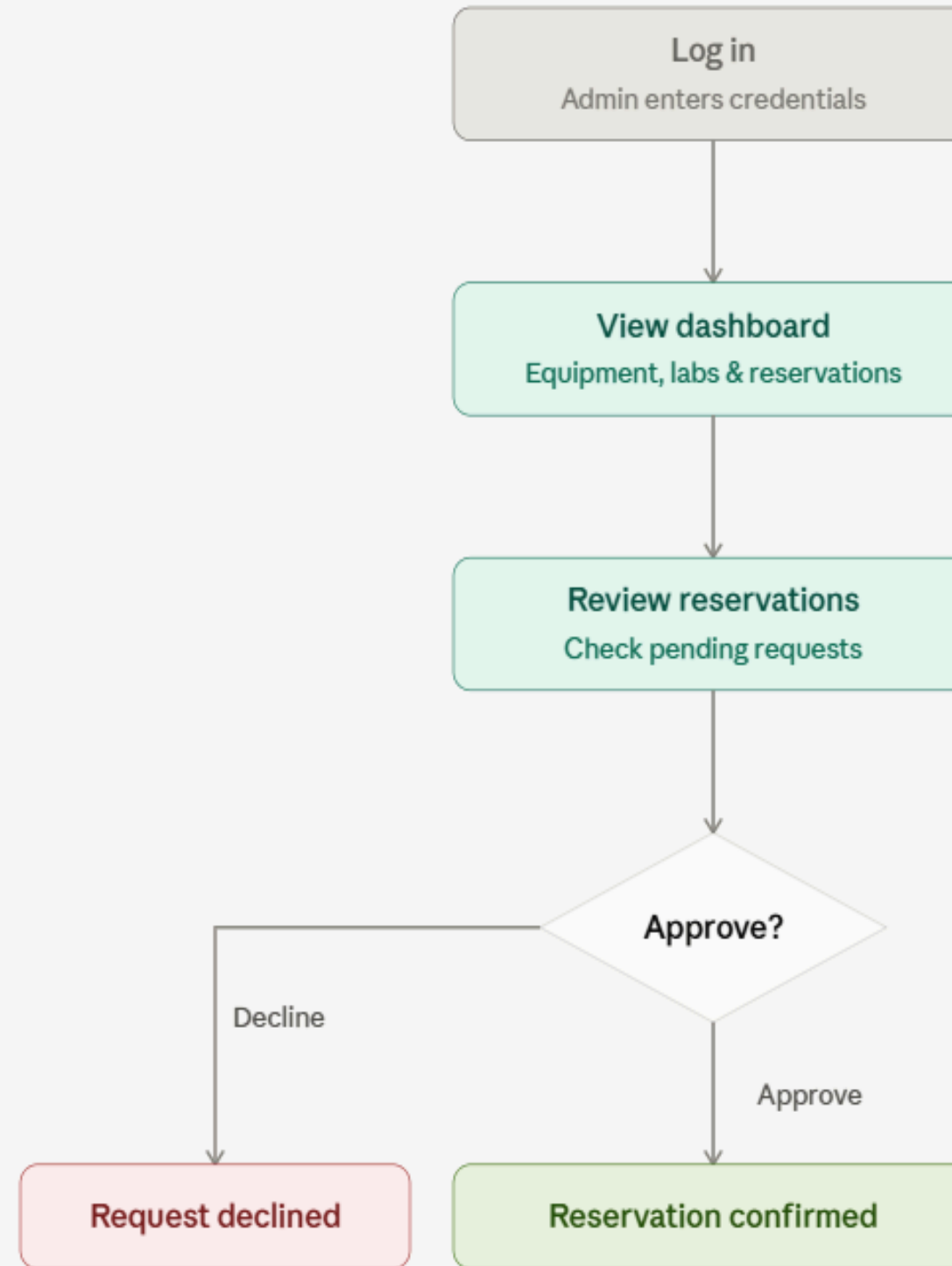
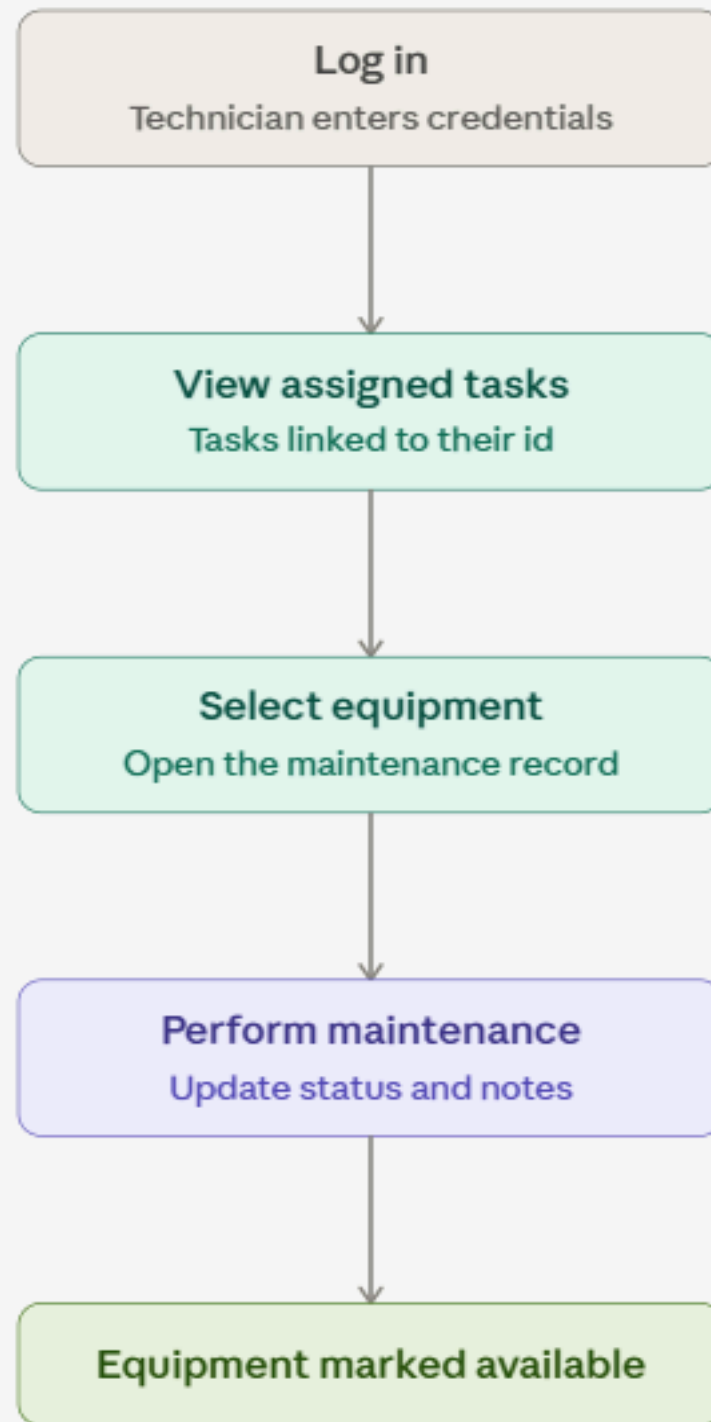
Relationships

1 One * Many

Relationships

Laboratory 1 → * Equipment
Equipment 1 → * Reservation
Equipment 1 → * Maintenance
Laboratory Staff 1 → * Reservation
Technician 1 → * Maintenance

UserFlow Diagram



isActive in CRUD Operations

Operation	Behavior	Example
Create	Record is added with isActive = true	New equipment is active and visible.
Read	Fetch only records where isActive = true	Active equipment list.
Update	Update allowed only if isActive = true	Cannot update deactivated records.
Delete	Set isActive = false (Soft Delete)	Record is hidden but still in database.

Benefits: Keeps data history, can be restored, better for auditing.

isActive — Soft Delete

Deactivating records instead of deleting them

What is isActive?

Instead of deleting records permanently from the database, the system uses the isActive field to deactivate a record.

BaseEntity

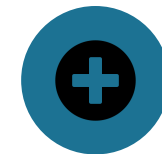
```
@MappedSuperclass
public abstract class BaseEntity {

    @Id
    @GeneratedValue(strategy = IDENTITY)
    private Integer id;

    private LocalDateTime createdAt;
    private LocalDateTime updatedAt;

    private Boolean isActive = true;
}
```

How It Works



Add / Create

isActive = true



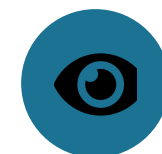
Update

Update fields



Delete

isActive = false (not physically deleted)



Get / List

Fetch only where isActive = true

API Endpoints

GET	/api/equipment	Get all equipment (active)
GET	/api/equipment/{id}	Get equipment by id
POST	/api/equipment	Add new equipment
PUT	/api/equipment/{id}	Update equipment
DELETE	/api/equipment/{id}	Deactivate equipment (isActive = false)
GET	/api/laboratories	Get all laboratories
GET	/api/reservations	Get all reservations
POST	/api/maintenance	Add maintenance record

Conclusion

The Healthcare Equipment System provides an efficient way to manage medical equipment, laboratories, reservations, and maintenance — with a scalable architecture, a soft-delete mechanism, and containerized deployment using Docker.



Reliable. Scalable. Maintainable.
Built for Healthcare.

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- ✓ Equipment Mgmt
- ✓ Laboratory Mgmt
- ✓ Reservations
- ✓ Maintenance
- ✓ Soft Delete
- ✓ REST API
- ✓ Docker Deploy

